

EXERCISES AND EXACT SOLUTIONS

Distance Is Not Access

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C24
Notebook	V7_B_N04
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Replace Euclidean with travel time.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Add facility capacity.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Define equity monitoring.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Use a validated routable network with mode, season, closures, and border constraints.

Solution 2

Allocate catchments under service-specific capacity and waiting-time constraints.

Solution 3

Publish population-weighted access distributions by protected and geographic groups with uncertainty and disclosure controls.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated